

BattleOps: Week 3

Theme: Kubernetes Security, Platform Hardening & Real-World Incident Prevention

Objective: Apply Kubernetes security concepts in real production-like situations, debug misconfigurations, enforce policies, and secure workloads end-to-end.

WEDNESDAY: Service Accounts, OIDC, Workload Identity

ADVANCED CONCEPTS

- How workloads authenticate to cloud services
- OIDC federation → IAM Roles → Pod identity
- Why storing cloud keys in Secrets is insecure

KEY OUTCOMES

- You can configure Workload Identity / IRSA
 - You can remove hardcoded secrets inside pods
 - You understand identity → permission → workload mapping
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WHAT TO PRACTICE

Lab 1: Create a Cloud-Aware Service Account

Create SA → annotate → attach IAM Role (IRSA/WI).

Lab 2: Deploy a Pod that Authenticates Without Keys

Call AWS/GCP APIs using pod identity.

Lab 3: Break & Fix Identity

Remove annotation → observe: AccessDenied

Reapply identity → validate success.

WHERE TO STUDY

- AWS IRSA docs
 - GCP Workload Identity docs
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INFRATHRONE BATTLEOPS NOTES

- Any pod with cloud keys is already compromised.
 - Identity-based access is the future of cloud security.
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